

Why Storage-Time Compression Fails

An Information-Theoretic Analysis of Deferred Projection in LLM Memory Systems

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Abstract

Recent work—E-mem, Yamanaka et al., and MemPalace—has independently converged on a shared architectural critique: compressing information at ingestion time, before queries are known, destroys context that query-time reconstruction could recover. This paper provides a systematic information-theoretic analysis of **why** storage-time compression is structurally suboptimal. Applying rate-distortion theory, time-bounded epiplexity, and the information bottleneck method to the memory architecture design problem, we derive a **Projection Loss Conjecture** that decomposes the deferred-projection advantage into three interacting factors: query entropy $H(P_Q)$, document structural diversity Δ_{struct} , and storage budget tightness $f(B/H)$. We prove a Necessity Lemma establishing that any valid lower bound must involve all three factors. A toy model (Appendix) validates the three-factor structure and the zero-product property. The analysis yields three testable predictions and explicit boundary conditions specifying when storage-time compression IS optimal. This paper does not propose a new architecture; it provides the theoretical framework that a rapidly converging but fragmented research area currently lacks.

Introduction

LLM-based knowledge management systems—RAG pipelines, note-taking tools, conversation summarizers—share a common premise: information should be processed into structured form at the point of storage. Documents are chunked, embedded, summarized, or extracted into knowledge graphs at ingestion, before any query is known.

Within a single twelve-month window (late 2025 to mid-2026), multiple independent research groups have published pointed critiques of this default.

E-mem [1] identifies “destructive de-contextualization” caused by compressing sequential dependencies into rigid structures.

Yamanaka et al. [2] explicitly contrast the dominant `extract then store` paradigm with `store then on-demand extract`—an approach we abbreviate as STONE (`{S}tore {T}hen {O}n-demand {E}xtract`) for the remainder of this paper.

MemPalace [3] demonstrates through verbatim storage benchmarks that zero-processing write paths achieve 96.6% R@5 on LongMemEval—driven primarily by off-the-shelf embeddings applied to raw text, with its spatial-metaphor architecture contributing negligibly [4].

These works establish an *empirical* case: storage-time compression causes measurable information loss, and query-time reconstruction can recover capability that compression destroys. What they do not provide is a *theoretical* account of why the problem is structural rather than contingent. Without such an account, storage-time compression can be dismissed as an implementation detail to be optimized—make better embeddings, train better chunkers—rather than recognized as a structural property of the problem setting.

This paper supplies that account. We apply three established information-theoretic frameworks—rate-distortion theory [10], time-bounded epiplexity [9], and the information bottleneck method [11]—to the memory architecture design problem. Our contribution is not in proving novel theorems, but in applying these frameworks to a domain where they explain a phenomenon that has been observed empirically but not yet understood theoretically.

Contributions. (1) A Projection Loss Conjecture that decomposes the deferred-projection advantage into three interacting factors (§3). (2) A Necessity Lemma proving that all three factors must appear in any valid lower bound (§3.4). (3) Three testable predictions with explicit boundary conditions (§4). (4) A toy model validating the three-factor structure and zero-product property (§Appendix).

Relation to a companion system release. The four design principles derived in §4 were not formulated in the abstract. They were extracted from a working prototype that the author has run continuously to support personal knowledge management workflows. This paper provides the theoretical accounting for that prototype’s design choices; a separate release will describe the system itself, its operating behavior, and qualitative observations from sustained use. The information-theoretic argument presented here stands independently of the prototype’s specific implementation: the predictions of §4 can be tested by any DPM-conformant system, and the boundary conditions of Table~2 apply equally to all instances of the class. Cross-validation between the theoretical predictions and the prototype’s empirical behavior is left to the companion

release and to future controlled experiments on standard benchmarks.

Related Work

Critical memory architectures. E-mem [1] proposes episodic context reconstruction with 7.75% F1 improvement and 70% token reduction on LoCoMo. STONE [2] advocates “store then on-demand extract.” MemPalace [3] provides verbatim storage benchmarks [4]. These works share the diagnosis; ours provides the theory.

Late interaction methods. ColBERT [5] establishes that per-token embeddings with max-similarity scoring outperform single-vector representations. Late Chunking [6] defers chunk boundaries to query time (+6.5 nDCG@10 on NFCorpus). LatentMAS [7] avoids premature projection to text in multi-agent systems (14.6% accuracy gain, 70–83% token reduction). These methods converge on “defer projection” from different angles; we extend the principle to memory architecture with information-theoretic justification.

Foundational theory. Power sampling [8] demonstrates that capabilities exist latently in model weights. Epiplexity [9] formalizes the information accessible to a time-bounded learner. The information bottleneck [11] characterizes optimal compression under relevance constraints. We apply all three to our analysis.

Memory benchmarks. LongMemEval [12] (ICLR 2025, 500 questions over multi-session chat) and LoCoMo [13] (ACL 2024) are the standard evaluation frameworks. Our experimental predictions target these benchmarks.

Information-Theoretic Analysis

Rate-Distortion Theory Applied to Memory Systems

Shannon’s rate-distortion function [10] gives the minimum bits required to represent source X with expected distortion $\leq D$:

$$R(D) = \inf_{P_{\hat{X}|X}: \mathbb{E}[d(X, \hat{X})] \leq D} I(X; \hat{X}) \quad (1)$$

The critical implicit assumption is that the distortion measure d is known to the encoder at compression time. For a memory system serving unknown future queries, this assumption fails. For each query $q \in \mathcal{Q}$, there exists an induced distortion d_q that penalizes reconstructions differently depending on what information q requires. A summary capturing algorithmic details but omitting performance characteristics is low-distortion for a correctness query but high-distortion for a latency query.

$$R_q(D) = \inf_{P_{\hat{X}|X}: \mathbb{E}[d_q(X, \hat{X})] \leq D} I(X; \hat{X}) \quad (2)$$

When $d_{q_1} \neq d_{q_2}$, the optimal compression for q_1 is generally not optimal for q_2 .

Observation 1 (Query-Conditional Advantage). Let P_Q be the query distribution. A storage-time system commits to a single $P_{\hat{X}|X}$ at ingestion, while a deferred system adapts per query:

$$\mathbb{E}_q \left[\inf_{P_{\hat{X}|X}^{(q)}: \mathbb{E}[d_q] \leq D} I(X; \hat{X}^{(q)}) \right] \leq \inf_{P_{\hat{X}|X}: \mathbb{E}_q[\mathbb{E}[d_q]] \leq D} I(X; \hat{X}) \quad (3)$$

with strict inequality when $d_{q_1} \neq d_{q_2}$ and documents contain information relevant to one query but not the other. *Justification:* set inclusion—the per-query-adaptive feasible set contains the fixed scheme’s feasible set.

Observation 1 is not a deep result—it is “more flexibility at decision time cannot hurt” restated in information-theoretic terms. Its value is making explicit *what conditions* cause the inequality to be strict: diverse queries and multi-aspect documents. Under a storage budget B , the deferred system pays a storage premium ($H(X)$ vs. B) for query-time flexibility. Whether this trade-off favors deferred projection depends on query entropy $H(P_Q)$, document structural diversity, and budget tightness—the three factors of Conjecture 1.

Time-Bounded Epiplexity

Following Finzi et al. [9], we separate information in data x for a time-bounded learner into structural epiplexity $S_T(x)$ (learnable within budget T) and time-bounded entropy $H_T(x)$ (noise-like to the bounded learner). Critical finding: deterministic transformations can shift information between these components:

$$S_T(\phi(x)) > S_T(x) \quad \text{is possible for deterministic } \phi \quad (4)$$

Chain-of-thought prompting is canonical: explicit intermediate steps increase structural epiplexity without adding new information. Equation~(4) establishes that deterministic transformations *can* increase S_T ; it does not establish that all deterministic transformations preserve or increase it. We next identify the specific class of transformations—content-reductive ones (chunking, summarization, single-vector embedding)—that demonstrably reduce S_T for at least some query-document pairs:

$$S_T(C(x)) < S_T(x) \quad (5)$$

for mainstream RAG, KG extraction, and summarization. Storage-time compression permanently reduces the structural epiplexity available for query-conditional extraction.¹

Observation 2 (Structural Preservation). A system storing x with near-identity transformation ($\hat{x} \approx x$) preserves $S_T(x)$ for all queries. Any non-invertible storage-time compression achieves $S_T(C(x)) < S_T(x)$ for some document-query pairs.

¹The epiplexity framework does not preclude *structure-revealing* storage-time augmentations (e.g., cross-reference detection) that add explicit structure without removing content. DPM’s near-identity storage is the simplest guarantee of S_T preservation, not the only one.

Information Bottleneck with Unknown Targets

The IB method [11] minimizes $I(X;T) - \beta \cdot I(T;Y)$ for a known target Y . In memory systems, Y is the answer to an unknown future query. Storage-time systems optimize for an assumed target Y_{assumed} ; deferred systems optimize for the actual Y_q at query time.

Observation 3 (Adaptive Bottleneck). For any fixed β , the deferred IB scheme achieves $\mathbb{E}_q[I(A;Y_q)] \geq I(\hat{X};Y_{\text{assumed}})$, with strict inequality when the optimal β_q varies across queries. *Justification:* the deferred scheme can replicate the storage-time scheme by setting $\beta_q = \beta$.

On the relationship among Observations 1–3. The three observations share a common set-inclusion structure: in each framework, a deferred scheme can replicate the corresponding storage-time scheme as a special case, so the deferred-feasible set contains the storage-feasible set. The three frameworks differ in *what is being optimized*—rate under a distortion criterion (RDT), structural epiplexity for a bounded learner, or predictive information under an IB tradeoff—not in the dominance mechanism. Their joint application clarifies which architectural property each criterion privileges (representation size, computational tractability, task relevance, respectively), rather than constituting three independent lines of evidence for a single conclusion.

The Projection Loss Conjecture

Observations 1–3 establish *weak dominance* but not magnitude. We now state a conjecture quantifying the deferred-projection advantage.

Conjecture 1 (Projection Loss). For a memory system with document distribution P_X , query distribution P_Q , storage budget B , and utility U , let $\mathcal{C}_{\text{storage}}$ be any storage-time compression within budget B , and $\mathcal{D}$ a deferred projection scheme storing documents near-losslessly. Then:

$$\mathbb{E}_{(x,q)} [U(\mathcal{D}) - U(\mathcal{C}_{\text{storage}})] \geq H(P_Q) \cdot \Delta_{\text{struct}} \cdot f\left(\frac{B}{H(X)}\right) \quad (6)$$

where:

- $H(P_Q)$ is query distribution entropy (nats). $H = 0$ when all queries are identical.
- $\Delta_{\text{struct}} = \mathbb{E}_x [\mathbb{E}_{q_1, q_2 \sim P_Q} [1 - \text{sim}(C_{q_1}^*(x), C_{q_2}^*(x))]]$ is the structural diversity of the collection—the expected pairwise dissimilarity, across query pairs, of the optimal extractions $C_q^*(x) = \arg \min_{C \in \mathcal{C}} d_q(x, C)$ from the same document. $\text{sim}(\cdot, \cdot)$ is any bounded pairwise similarity on the representation space $\mathcal{C}$ (e.g., cosine, normalized edit similarity, or LLM-as-judge consistency rating). $\Delta_{\text{struct}} = 0$ when every document requires the same extraction regardless of query; Δ_{struct} approaches its maximum when different queries demand near-disjoint extractions. The pairwise formulation avoids requiring $\mathcal{C}$ to admit a well-defined mean and accommodates non-vector representations (text, graphs) on equal footing.
- $f(B/H(X))$ is a budget sensitivity function, monotonically decreasing, with $f(0) = 1$, $f(\infty) = 0$, $f' < 0$. The hypothesized form is $f(x) = 1 - e^{-x}$ (diminishing marginal returns); alternatives include $f(x) = 1/(1+x)$ and linear threshold forms, distinguishable empirically.

On the units of Equation (6). The left-hand side is in task-specific utility units; the right-hand side combines nats with a similarity-metric variance and a dimensionless budget term. Equation (6) should therefore be read as a *structural* relation rather than a numerical inequality: it asserts that the projection-loss advantage is governed by the three factors named on the right and vanishes whenever any of them vanishes (Lemma~1). Operationalization for any specific deployment requires calibrating the unit reconciliation between $H(P_Q) \cdot \Delta_{\text{struct}}$ and U as an empirical choice, not a theoretical constant. The testable content of Conjecture~1 is the directional dependence captured in the three predictions of §4, which hold under this conceptual reading regardless of the specific calibration.

Conjecture~1 is a conjecture, not a theorem. A complete proof requires specifying the structural representation space, the similarity metric, and the mapping from structural fidelity to utility—specifications that depend on the specific LLM and task domain.

Necessity Lemma

While Conjecture 1 remains unproven in its full quantitative form, we can establish the structural necessity of the three factors.

Lemma 1 (Necessity). The projection loss $L = \mathbb{E}_{(x,q)}[U(\mathcal{D}) - U(\mathcal{C}_{\text{storage}})]$ satisfies:

1. $L = 0$ when $H(P_Q) = 0$ (all queries identical—storage-time can tailor compression)
2. $L = 0$ when $\Delta_{\text{struct}} = 0$ (uniform documents—single scheme suffices)
3. $L = 0$ when $B \geq H(X)$ (lossless storage feasible—no information discarded)

Proof sketch. (i) A single query q_0 with $\Pr[q_0] = 1$: optimize $P_{\hat{X}|X}$ for d_{q_0} . (ii) $\Delta = 0 \implies$ pairwise similarity $\text{sim}(C_{q_1}^*, C_{q_2}^*) = 1$ for all query pairs, so a single extraction scheme serves all queries equally, yielding zero utility gap. (iii) $B \geq H(X)$ enables $\hat{X} \approx X$, preserving all information. $\square$

Consequently, any lower bound on L must involve all three factors. The multiplicative form $H \cdot \Delta \cdot f$ is the minimal structure satisfying the zero-product property: $0 \cdot \Delta \cdot f = H \cdot 0 \cdot f = H \cdot \Delta \cdot 0 = 0$. Additive forms $(H + \Delta + f)$ would remain positive when one factor is zero, contradicting Lemma 1. Alternatives such as $\min(H, \Delta, f)$ satisfy the zero-product property but lose the amplification intuition; empirical discrimination between functional forms is left to future work.

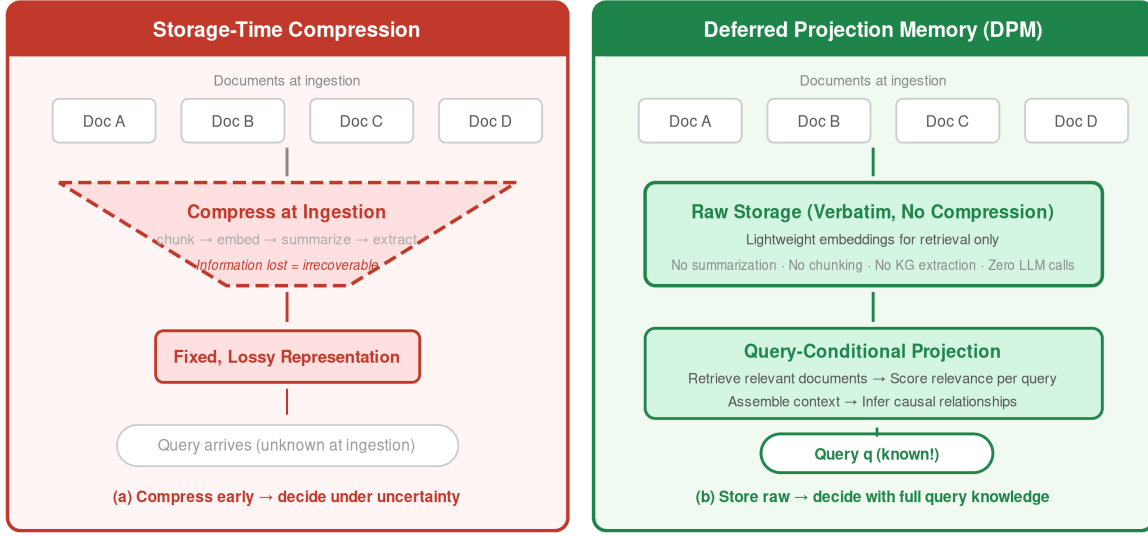


Figure 1: Storage-time compression (left) locks in a lossy representation before the query is known. Deferred projection (right) preserves raw content and applies query-conditional extraction.

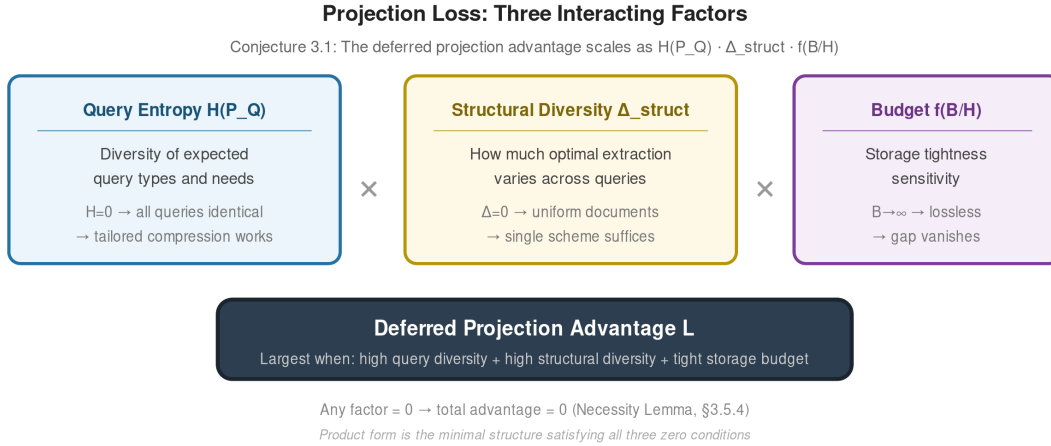


Figure 2: The three factors of Projection Loss. The product form satisfies the zero-product property: if any factor vanishes, the total advantage vanishes (Lemma 1).

Architectural Implications

The analysis yields four design principles for deferred-projection memory systems:

1. **Storage Preservation.** Store original text verbatim; no chunking for semantic boundaries, no embedding-only representation, no summarization (Observation 2).
2. **Query-Conditional Extraction.** Retrieve and project based on observed query, not at ingestion (Observation 1).
3. **Graded Relevance.** Return continuous confidence scores $c_i \in [0, 1]$ rather than binary relevant/irrelevant judgments, enabling evidence weighting (Observation 3).

4. **Lazy Relationship Inference.** Infer document relationships at query time, conditioned on the specific query, rather than pre-committing to fixed schemas (Equation 4).

These principles collectively define the Deferred Projection Memory (DPM) architectural class, of which E-mem, STONE, and MemPalace are instances at varying levels of conformance. Table~1 compares DPM against dominant paradigms.

Table 1: Architectural comparison.

Property	RAG	Knowledge Graphs	DPM
Storage	Chunked + embedded	Pre-extracted triples	Verbatim
Write cost	50–200 ms/doc	100–500 ms/doc	10–50 ms/doc
Relevance	Binary (top-k)	Binary (edge exists)	Continuous $c_i \in [0, 1]$
Causality	Coincidental co-retrieval	Fixed at ingestion	Query-conditional
Retention	Lossy	Lossy	Lossless

Table 2: When deferred projection provides advantage vs. when alternatives are preferable.

Condition	Favors Deferred Projection	Favors Alternatives
Query diversity	High	Low (homogeneous, predictable)
Document heterogeneity	High (mixed formats)	Low (uniform type)
Storage budget	Constrained	Abundant (all fits in context)
Latency	>2s acceptable	Sub-500ms required
Cross-doc reasoning	Required (causal chains)	Not needed

Empirical Predictions

Conjecture 1 yields three testable predictions. Each is a statistical hypothesis that does not depend on the conjecture’s formal status.

Prediction 1 (Query Distribution). The relative advantage of deferred projection increases with query entropy $H(P_Q)$. Testing: compare ΔAQ between homogeneous (single query type) and heterogeneous (mixed types) query sets.

Prediction 2 (Structural Diversity). The advantage is larger for document collections with high Δ_{struct} —mixed code, documentation, discussions—than for uniform collections.

Prediction 3 (Budget Sensitivity). The advantage increases as storage budget tightens. Under abundant budget (entire collection fits in context window), deferred projection and direct injection perform comparably ($f(\infty) = 0$). Under tight constraints (artificially limited context), the advantage is maximized.

Falsification criteria. The framework is falsifiable: (a) if no query-type interaction is observed, the

query-conditionality premise is disconfirmed; (b) if direct injection matches deferred projection on quality at comparable cost even under tight budget, retrieval-based projection provides no practical benefit; (c) if BM25 lexical matching achieves comparable causal chain completeness, deferred projection’s semantic components add cost without value.

Power analysis. For the minimum viable experiment (150 documents, $n = 25$ per query category), detecting $\delta = 0.5$ on a 5-point LLM-as-judge scale (assumed $\sigma \approx 0.8$) requires $n \approx 42$ per group at $\alpha = 0.05$, $1 - \beta = 0.80$; $n = 25$ yields power ≈ 0.55 . The experiment is powered only for large effects ($\delta \geq 0.75$). Medium but non-significant effects should be interpreted as inconclusive.

Discussion

Limitations. (1) The rate-distortion framing assumes well-defined d_q , but distortion in LLM systems is mediated by complex reasoning. (2) Δ_{struct} remains unoperationalized—plausible proxies include embedding-space variance, edit distances between extracted fact sets, and LLM-as-judge similarity. (3) Observations 1–3 establish weak dominance, not magnitude. (4) The product decomposition in Conjecture~1 is the simplest form consistent with Lemma~1 but not the only one; interaction terms, regime-dependent scaling, or alternative parameterizations may better capture empirical behavior.

Relationship to Long-Context Evolution. Expanding context windows are the most serious challenge to deferred projection’s relevance. A simple cost analysis suggests that retrieval becomes preferable for collections beyond a small threshold—on the order of tens of documents at 2026 frontier-model pricing—above which direct injection’s per-query token cost dominates retrieval’s amortized cost. A more precise breakeven depends on assumed retrieval call cost, average document length, top- k value, and per-token model pricing, and is sensitive to which model and provider are assumed. Even when context windows make retrieval optional, the attention efficiency argument remains: LLM attention is a finite resource, and filtering to query-relevant content concentrates that resource where it matters.

Open problems. (1) Efficient estimation of the parametric-external knowledge boundary. (2) Compute-aware budget allocation across retrieval, assembly, and reasoning. (3) Longitudinal validation: do deferred-projection users make better decisions over months?

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Appendix: A Toy Model of Projection Loss

We construct a fully-specified combinatorial model where the projection loss admits closed-form analysis, validating the three-factor structure and zero-product property.

Model Definition

Let there be n atomic facts, each $x_i \sim \text{Bernoulli}(1/2)$. A document $x \in \{0, 1\}^n$ is the complete vector. There are T query types; type t requests fact subset $S_t \subset \{1, \dots, n\}$ of size k . P_Q is uniform: $H(P_Q) = \log T$. Storage budget: $m < n$ bits. Utility: fraction of requested facts correctly retrieved. Incidence matrix $C \in \{0, 1\}^{n \times T}$ where $C_{i,t} = 1$ if query t needs fact i ; row sums $r_i = \sum_t C_{i,t}$.

Optimal Strategies

Storage-time stores the m facts with largest r_i : $\mathbb{E}[U_{\text{storage}}] = \frac{1}{Tk} \sum_{i \in M} r_i$. Deferred stores all n bits: $\mathbb{E}[U_{\text{deferred}}] = 1$. Loss: $L = 1 - \frac{1}{Tk} \sum_{i \in M} r_i$.

Three Canonical Cases

Case I ($T = 1, H = 0$): all queries identical. Storage-time stores exactly those k facts. $L = 0$. Case II (r_i constant, $\Delta = 0$): every fact equally important. $L = 1 - m/n$, depending only on budget. Case III (disjoint queries, $\Delta = 1, m \ll Tk$): $L \geq 1 - \lfloor m/k \rfloor / T \rightarrow 1$ for large T . Loss maximized when all three factors are at extremes.

Lower Bound

The projection loss satisfies $L \geq \Delta \cdot \max(0, 1 - m/(n(1 - \Delta)))$, satisfying the zero-product property. The query entropy $H = \log T$ enters through Δ : larger T enables larger Δ , but the relationship is mediated by incidence matrix structure.

Relationship to Conjecture 1

The toy model validates: (i) all three factors are necessary, (ii) the zero-product property holds, (iii) effect directions match ($H \uparrow, \Delta \uparrow, m/n \downarrow$ all increase L). It does not produce exact multiplicative factorization $L = H \cdot \Delta \cdot f - T$ and Δ interact through the incidence matrix in ways that resist clean separation. The multiplicative form in Conjecture 1 is the simplest functional form consistent with all three structural properties, not a claim of exact factorization.